With the discovery of their level error by GIT, the Riemann and Collatz conjectures are confirmed in their own structure.

Abstract

The Collatz conjecture and the Riemann hypothesis are among the best-known unsolved problems in mathematics. They have occupied researchers for decades, tying up immense computing resources and serving as touchstones for the mathematical understanding of finiteness and infinity. This work shows that both problems are based on a common **level error**: the mixing of the finite processes of space-time with the infinite potential of dimension 0. With the help of **Holistic Information Theory (HIT)**, a clear separation of these levels is made. As a result, the Collatz and Riemann conjectures appear not as unsolvable puzzles, but as manifestations of universal laws of information. In this sense, both conjectures are confirmed by an error in their classical formulation. In addition, it is shown that information, consciousness, mathematics and geometry represent independent laws of information, the interconnection of which forms the foundation of the evolution of nature, mind and the universe.

Keywords

Collatz conjecture; Riemann hypothesis; holistic information theory; dimension 0; nothingness; level error; law of information; geometry; consciousness; natural law of information

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1. Introduction

Mathematics thrives on open questions. Some of them are technical problems that can be solved by clever methods, new proofs or the use of modern computer technology. Others, however, are so deeply rooted that they remain open for generations and become the basis for entire branches of research.

Two particularly prominent examples are the **Collatz conjecture** (1937) and the **Riemann hypothesis** (1859). Both have something paradoxical about them: they are easy to formulate, yet unsolvable within the classical framework. They therefore fascinate not only experts, but also laypeople, as they exemplify how mathematics reaches the limits of its thinking.

The Collatz conjecture initially seems like a mathematical game. Choose any natural number, halve it if it is even, or multiply it by three and add one if it is odd. Repeat the process. The conjecture: no matter which number you choose, the sequence always ends in the cycle $4 \to 2 \to 14 \to 2 \to 14 \to 2 \to 1$. A rule that can be explained in a minute has eluded proof for over 80 years.

The Riemann hypothesis, on the other hand, is considered one of the "holy grails" of mathematics. It concerns the distribution of prime numbers and states that all non-trivial zeros of the zeta function lie on the line ℜ(s)=1/2\Re(s) = 1/2ℜ(s)=1/2. Its truth or falsity would have fundamental consequences – for number theory, for cryptography, for the modelling of random processes and possibly even for physics.

Both problems seem fundamentally different, but they share one thing in common: their insolubility does not result from a lack of intelligence or technology, but from a **level error**. Classical mathematics mixes the level of **space-time** – in which only finite processes are realised – with the level of **dimension 0**, in which infinite possibilities reside.

Holistic Information Theory (HIT) offers a new approach here. It makes a strict distinction between:

- **Dimension 0**: infinite potential, timeless and spaceless, source of all information.
- **Nothingness**: mediator that connects questions from space-time with answers from dimension 0.
- **Space-time**: realm of finite realisations in which processes take place.

When these levels are mixed, seemingly unsolvable puzzles arise. If they are neatly separated, the Collatz and Riemann conjectures become manifestations of universal laws of information.

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2. The Collatz conjecture – iterations between finiteness and potential

2.1 Classical formulation

The Collatz conjecture defines a simple iteration:

- n↦ n/2n \mapsto n/2n↦ n/2, if nnn is even.
- n↦ 3n+1n \mapsto 3n+1n↦ 3n+1, if nnn is odd.

Claim: Every sequence reaches the cycle 4→2→14 → 2 → 14→2→1.

Its fascination lies in the fact that it is so easy to understand that even a schoolchild can comprehend it – and yet no proof exists. To date, billions upon billions of starting numbers have been tested without a single exception ever being found. However, the possibility of infinite trajectories could not be ruled out in classical thinking.

2.2 Historical efforts

The Collatz conjecture has been the subject of intensive research since the 1970s. John Conway showed that similar iterations can be undecidable, suggesting that Collatz lies at the edge of what is provable. Terras and Crandall developed heuristic models that interpret Collatz sequences as a kind of "random process".

Paul Erdős summed it up: *"Mathematics may not be ready for such problems."* Finally, in 2019, Terence Tao proved that almost all Collatz orbits are "almost bounded." This provided strong confirmation, but not a complete proof.

2.3 GIT view: finiteness instead of infinity

GIT shows that only finite processes exist in spacetime. The idea of an "infinite Collatz orbit" is a category error. Infinity belongs exclusively in dimension 0 as potential, not as a realised orbit.

It follows that every Collatz sequence necessarily enters a finite cycle. The only cycle that exists stably is 4→2→14 \to 2 \to 14→2→1.

2.3a Numbers as markers of finiteness

Every number is a **limitation**. It marks a "this much and no more" and thus belongs to the plane of space-time. An "infinite number" is a contradiction.

The infinity of the number series does not lie in the numbers themselves, but in the potential for unlimited continuation – that is, in dimension 0. Every number is finite, and every operation produces a finite number again.

It follows that an infinite path does not exist in space-time, but only as an imagined potential. Therefore, all sequences end in the cycle 4→2→14 \to 2 \to 14→2→1.

2.4 Information potential as an explanation

Collatz describes GIT with an information potential function Φ\PhiΦ. Each iteration step changes the information density: even steps reduce it, odd steps increase it in the short term, but on average it decreases.

The super-nothing conveys only finite realisations. Infinite trajectories contradict the laws of information. Thus, Collatz is not only confirmed by GIT, but also explained.

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3. The Riemann hypothesis – prime number symmetry and information equilibrium

3.1 Classical formulation

The zeta function is:

ζ(s)=∑n=1∞1ns,ℜ(s)>1.\zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s}, \quad \Re(s) > 1.ζ(s)=n=1∑∞ns1,ℜ(s)>1.

It can be continued analytically. The hypothesis: All non-trivial zeroes lie on the line ℜ(s)=1/2\Re(s)=1/2ℜ(s)=1/2.

3.2 Significance for mathematics

The Riemann hypothesis is more than just a number theory problem. It concerns:

- **Prime number theorem**: Accuracy of the estimation of prime numbers.
- **Cryptography**: The security of modern encryption is based on prime number symmetries.
- **Statistics & quantum mechanics**: Montgomery's pair correlation shows parallels to eigenvalues of random matrices.
- **Cosmology**: Some models interpret zeroes as symmetries of energy states.

A proof or counterproof would have fundamental consequences.

3.2a Prime numbers between finiteness and infinity

Prime numbers are also finite. Each one is a marker in space-time. The "infinity of prime numbers" simply means that the process of their creation never ends.

The plane separation of the GIT explains:

- **Dimension 0** contains the inexhaustible potential of new prime numbers.
- The **super-nothing** mediates their orderly insertion.
- Every single prime number appears in **space-time**.

The infinity of the prime number series is an expression of dimension 0, not a property of the numbers themselves. The Riemann hypothesis marks the balance in the straight line $\Re(s)=1/2$\Re(s)=1/2$\Re(s)=1/2$.

3.3 GIT view: Prime numbers as elementary information

Prime numbers are elementary units of information. Composite numbers are networks. The nothingness ensures balance: every new piece of information fits in symmetrically. This symmetry appears in the critical straight line.

3.4 Operatorial formulation

GIT describes this using a self-adjoint information operator HHH. Its spectrum is real, so the zero points lie on $\Re(s)=1/2$\Re(s)=1/2$\Re(s)=1/2$. This is how GIT explains the Riemann hypothesis as an expression of universal information laws.

4. Philosophers, plane errors and the role of self-awareness

The question of finiteness and infinity has been one of the fundamental themes of philosophy since ancient times. Even early thinkers recognised that although the human mind thinks in

finite categories, it is inevitably confronted with the concept of infinity. The Collatz and Riemann conjectures are modern mathematical variants of this age-old tension. They show that the problem cannot be solved by formal proofs alone as long as the planes are not clearly distinguished.

Heraclitus summed up the becoming of the world in his famous phrase "Everything flows". He saw reality as a process, as constant change – an anticipation of the space-time level, in which everything finite is subject to change. His opponent Parmenides, on the other hand, declared being to be unchanging and timeless. In doing so, he implicitly described dimension 0, the sphere of infinity that underlies all change. The contradiction between the two positions was only later systematically formulated by Plato.

Plato distinguished between the world of ideas and the world of appearances. Ideas are eternal, unchanging and perfect; appearances, on the other hand, are transient and imperfect. Here, too, we find the distinction between dimension 0 (the world of ideas) and space-time (the sensory world), but without a mediating principle. Aristotle filled this missing link with his doctrine of potentia and actus: between possibility and realisation there is a process that connects the two. This is precisely where GIT comes in with its concept of the super-nothing: a coupling field that connects questions of space-time with answers from dimension 0.

Plotinus later took up these ideas in his doctrine of the One. The One is the origin of all things, beyond space and time. Everything that exists emanated from this One without diminishing it. In modern language, one could say: the One corresponds to the infinite reservoir of information in dimension 0. Kant, in turn, formulated the antinomies of pure reason: the universe must both have a beginning and no beginning; it must be both finite and infinite. Kant thus reveals the irresolvable contradictions that arise when dimension 0 and space-time are mixed.

Hegel finally attempted to resolve these antinomies dialectically. For him, the logic of contradictions was a way to attain higher knowledge. But without explicitly separating the levels, his solutions often remained abstract. Only GIT offers a precise model: dimension 0 (potential), supernothing (coupling) and spacetime (realisation). This explains why seemingly unsolvable problems such as Collatz and Riemann arise – they are products of a level error.

Heidegger brought the discussion into the 20th century by emphasising finitude as a fundamental condition of human existence. Man is "being towards death" – his finitude structures all thinking. But it is precisely this finitude that opens up the horizon to the infinite. In GIT language, this means: only because consciousness is limited in space-time can it ask questions that open up the beyond and receive answers from dimension 0.

Self-awareness plays a decisive role here. It is the "questioner" who activates the connection between finiteness and infinity in the first place. Without consciousness, there would be no questions and thus no flow of information from dimension 0 into space-time. From this perspective, philosophy and mathematics are not mere systems of thought, but expressions of a deeper law of information.

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5. Comparison: Collatz and Riemann as plane errors

At first glance, the Collatz conjecture and the Riemann hypothesis appear to be completely different. One is playful, the other of great depth. But both share the same error in thinking: the mixing of finiteness and infinity.

In Collatz's case, the error lies in treating infinite paths as a real possibility. Classically speaking, a number could generate an infinite sequence without a cycle. But every number is finite, every step is finite, and space-time knows no infinite processes. The possibility of an infinite trajectory belongs exclusively to dimension 0. Once this separation is recognised, it becomes clear that all sequences must end in a finite cycle. The "proof" is therefore not a complicated calculation, but a consistent separation of planes.

Riemann's hypothesis is more subtle. Here, the error lies not in a supposed infinity of processes, but in the interpretation of prime number symmetries as a purely analytical problem without informational content. Classical number theory regards the zeta function as an abstract object whose zeroes are to be determined purely formally. But GIT shows that prime numbers are elementary units of information, and their symmetry is an expression of a deeper order of information. The critical line $\Re(s)=1/2\Re(s)=1/2\Re(s)=1/2$ marks the balance between potential (dimension 0) and realisation (space-time).

In both cases, it becomes clear that the level error creates the impression of insolubility. In Collatz's case, it is the false assumption of infinite realisations; in Riemann's case, it is the false abstraction without reference to information. If the levels are separated, both conjectures are confirmed. They are not unsolved, but misled by their classical formulation.

This insight has far-reaching consequences. It shows that mathematics is not just a symbolic language, but a natural law of information that must always be thought of in three levels: dimension 0, supernothing, space-time. Collatz and Riemann are examples of how seemingly insurmountable problems disappear when this structure is taken seriously. This makes them evidence of GIT and signposts to a new mathematical philosophy.

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6. Conclusion

The analysis of the Collatz and Riemann conjectures in the light of Holistic Information Theory shows that both problems are manifestations of the same error. They confuse finiteness and infinity, thereby creating paradoxes. The Collatz conjecture seems unsolvable because infinite paths are considered realistically possible. The Riemann hypothesis seems mysterious because the symmetry of prime numbers is not recognised as a law of information.

If we separate the levels, it becomes clear that in space-time there is only finiteness. Every number is a boundary, every step a finite realisation. Infinity belongs exclusively to dimension 0. The super-nothing mediates the exchange between the two spheres. In this sense, mathematics is not merely a tool, but a law of nature that structures the evolution of the universe, life and consciousness.

The conclusion is therefore that both assumptions are confirmed by an error in the assumption. Their insolubility is an apparent problem resulting from an incorrect level

assignment. GIT reveals that mathematics, consciousness, information and geometry are closely interlinked natural laws that together determine the dynamics of creation and evolution.

This view opens up new avenues. It shows that unsolved problems in mathematics are not only technical challenges, but also clues to deeper structures. By learning to separate planes cleanly, we can not only understand mathematical puzzles, but also gain new insights into philosophy, natural science and cosmology.

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7. Information, consciousness, mathematics and geometry as interconnected natural laws

1. **Traditional perspective**
   In classical science, mathematics and geometry are regarded as formal tools, not as laws of nature. Consciousness is seen as an emergent phenomenon, and information, while relevant in physics (Shannon, Landauer, quantum information), is not considered a "law of nature" in the sense of gravitation or thermodynamics.
2. **Extension**
   These four fundamental magnitudes are not merely tools or phenomena but **fundamental laws of nature** that together form an **information law**. This law is the basis of haemologous evolution, a universal evolution that encompasses not only biological but also cosmic, mental, and social processes.
3. **Integration into GIT**

- Mathematics = structure of information
- Geometry = spatio-temporal order of information
- Consciousness = reflection and creation of new information
- Information = elementary substance of energy, matter, and life

From this GIT perspective, an **information law** emerges that is not just one among many, but the **“meta-law”**, from which the other information laws can be derived.

GIT shows that information appears in many modalities: visual, acoustic, haptic, linguistic, mathematical, geometric, energetic. But all these forms are expressions of the same law. Information is not a human construct, but a universal structure.

Consciousness is the questioner that activates this information. It asks the questions that open up the beyond. Answers then emerge from dimension 0 into space-time. Mathematics clarifies the questions, geometry makes the structures visible. This creates an interconnected system: information, consciousness, mathematics and geometry form a whole.

Cosmologically, this means that the universe is the greatest possible finiteness. It grows by constantly feeding from dimension 0. Every new structure, every new insight is a realisation of infinite potential. Thus, the universe is not a closed system, but an open field of continuous creation.

The interconnection of these four laws – information, consciousness, mathematics, geometry – explains why Collatz and Riemann are not isolated puzzles, but expressions of the same structure. They show that even the most abstract problems of number theory are ultimately part of a universal process that connects the cosmos, mind and nature.

Literature

- Riemann, B. (1859). *On the number of prime numbers below a given magnitude*. Monthly Reports of the Berlin Academy.
- Collatz, L. (1937). *Integer functions*. Annual Report of the DMV.
- Kant, I. (1781). *Critique of Pure Reason*.
- Hegel, G. W. F. (1812–1816). *Science of Logic*.
- Heraclitus, Fragments (ca. 500 BC).
- Tao, T. (2019). *Almost all Collatz orbits attain almost bounded values*.

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8. Studies in relation to dimension 0

The theoretical considerations of GIT resonate in current studies dealing with finiteness and infinity. Ten examples illustrate this:

1. **Linnebo & Shapiro (2019):** Clearly distinguish between potential and actual infinity. → Confirmation of the separation of dimension 0 (potential) and spacetime (realisation).
2. **Lev (2024):** Shows that finite mathematics is more fundamental than infinite models. → Proves that infinity is only a borderline case from D0.
3. **Mageed (2025):** Emphasises the philosophical problems of infinity. → Without D0 separation, infinity remains a paradox.
4. **Ricœur (2022):** Finitude as a condition, infinity as a horizon. → Congruent with space-time vs. dimension 0.
5. **Cassirer & Heidegger (2022):** Infinity as a potential for thought, not as a real quantity. → Corresponds to the GIT view of D0.
6. **Baker (2024):** Finitude is constitutive, infinity remains a horizon. → Supports the GIT separation.
7. **Zeno analysis (2024):** Paradoxes arise from incorrect mixing. → Exactly what GIT describes as a level error.
8. **Routledge Encyclopedia:** Human finitude structures access to infinity. → Corresponds to the role of consciousness as questioner.
9. **Continental Philosophy Review (2006):** Philosophy is made possible by finitude. → Confirms the GIT thesis of consciousness as mediator.
10. **Leibniz Study (2021):** Infinities as mathematical fictions. → Proves that infinity is not realised, but only exists as potential in D0.

These studies show that GIT does not stand alone, but is embedded in a long tradition of philosophical and mathematical debates. They provide empirical and theoretical support for the thesis that Collatz and Riemann are explained and confirmed by a plane error.